

Introduction to Algorithms

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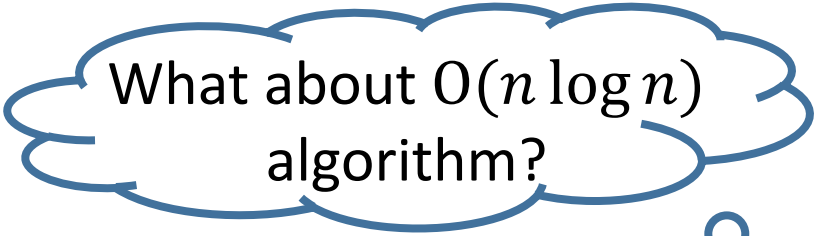
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18×10^9	112	34	21	For same computer, 5 CPU hours, which input size can be reached?
18×10^{10}	178	37	23	For 10 times faster computer, 5 CPU hours, which input size can be reached?

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Definition:

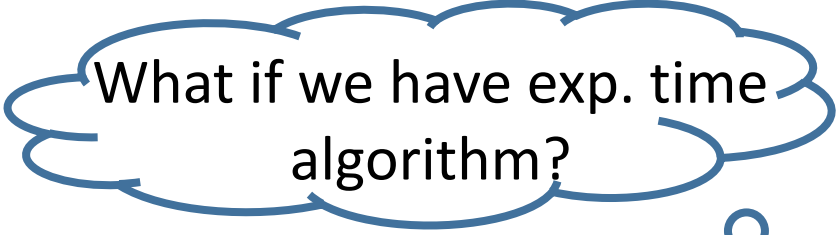
An algorithm is called *tractable* if its complexity is $O(n^k)$, where n is the input size and k is a constant.



What about $O(n \log n)$ algorithm?

Definition:

A problem is called *tractable* if it is solvable by a tractable algorithm.



What if we have exp. time algorithm?